

Library Data Warehouse

Transformation Rules Documentation

1. Introduction

This document describes the **data transformation rules** applied during the ETL process for the Library Data Warehouse.

The transformations are implemented in `etl_script.py` and focus on:

- Data standardization
- Advanced NULL handling
- Business rule enforcement
- Data quality improvement
- Consistent dimensional modeling

2. Source Systems

Source Dataset	Format	Description
Book Transactions	CSV	Physical book checkout and return data
Digital Downloads	Excel	E-book, journal, and article usage
Room Bookings	CSV	Study room reservations

3. General Transformation Principles

1. **Preserve Business Meaning of NULLs**
2. **Impute values only when statistically or logically valid**
3. **Flag ambiguous or suspicious data instead of deleting it**
4. **Standardize values before loading into the data warehouse**
5. **Track all transformations for audit and reporting**

4. NULL Handling Transformation Rules

4.1 Return Date (Book Transactions)

Attribute	Rule
return_date	If NULL → book is still checked out
Action	FLAGGED
Stored Value	NULL
Business Meaning	NOT_RETURNED

Reason:

A NULL return date indicates an active loan, not missing data.

4.2 Duration Minutes (Digital Downloads)

Attribute	Rule
duration_minutes	If NULL → impute average by resource type
Action	IMPUTED

Imputation Values:

Resource Type	Imputed Minutes
E-book	45
Journal	30
Article	20
Other	35

4.3 Student ID

Context	Rule	Action
Room Booking	NULL → walk-in user	FLAGGED (WALK_IN)
Other Transactions	NULL → data error	FLAGGED (DATA_ERROR)

Stored Value: UNKNOWN

4.4 Department

Condition	Rule
NULL + StudentID exists	Lookup from other records
Lookup fails	Set to Unknown

| Action | IMPUTED or FLAGGED |

4.5 Checkout Duration (Books)

Condition	Rule
Return date exists	Calculate duration
Return date NULL	Keep NULL

Calculation:

$\text{checkout_duration_days} = \text{return_date} - \text{checkout_date}$

5. Date Standardization Rules

5.1 Supported Input Formats

- YYYY-MM-DD
- MM/DD/YYYY
- DD-MM-YYYY

- YYYY-DD-MM
- Mon DD, YYYY
- DD-Mon-YYYY

5.2 Output Format

Field	Standard Format
All Dates	YYYY-MM-DD

Invalid dates are **flagged and logged**.

6. Categorical Standardization Rules

6.1 Department Names

Input Variations	Standardized Value
CS, CompSci	Computer Science
ENG, Engr	Engineering
BUS	Business
MED	Medicine
LAW	Law
ARTS	Arts
NULL / Empty	Unknown

6.2 Room Number

Input	Output
Room-101	R101

Input	Output
r 202	R202
ROOM303	R303
NULL	UNKNOWN

6.3 Time Slot

Keywords Detected	Standard Value
AM, Morning, 8–11	Morning
PM, 12–16	Afternoon
17–20	Evening
21–23	Night
Unknown	Morning (default)

7. Duplicate Handling Rules

Table	Duplicate Key	Rule
staging_book_transactions	transaction_id	Keep earliest record
staging_room_bookings	booking_id	Keep earliest record

Implementation:

Duplicates are removed using staging_id.

8. Dimension Transformation Rules

8.1 Student Dimension

Rule
Only CLEANED records are loaded
Missing department → Unknown
Walk-in users excluded
Slowly changing via effective_date

8.2 Resource Dimension

Source	Resource Type
Books	Physical Book
Digital	E-book / Journal / Article

Generated IDs:

DIGITAL_<resource_type>_<faculty>

8.3 Location Dimension

Attribute	Rule
Floor	Derived from room number
Capacity	Default = 4
Building	Main Library

9. Fact Table Transformation Rules

9.1 Book Checkout Facts

Attribute	Rule
date_key	Derived from checkout_date
is_returned	TRUE if return_date exists
duration	NULL if not returned

9.2 Room Booking Facts

Attribute	Rule
walk-in	TRUE if student_id = UNKNOWN
student_key	Default = 1 for walk-ins
resource_key	Fixed (room usage)

10. Data Quality Flagging

All transformed records receive:

data_quality_flag = 'CLEANED'